

4-Year Strategic Research Roadmap

Real-Time Aerodynamic State Estimation Using Surface Sensors and Data Assimilation

Overall Vision

Develop and experimentally validate a real-time data assimilation framework capable of predicting separated and dynamically stalled aerodynamic flows using only surface pressure/temperature sensors. The project culminates in a high-speed wind tunnel demonstration relevant to aerospace applications such as maneuvering aircraft, aeroelastic stability, and flow control.

PHASE 1 (Year 1): Foundational Development and Canonical Validation

Goals

- Build the complete data assimilation (DA) pipeline end-to-end.
- Validate the approach on controlled, canonical flows before moving to complex airfoil dynamics.

Key Actions

1. Develop Minimal Viable DA Framework

- Implement ROM and EnKF/UKF/hybrid filter.
- Demonstrate real-time update capability using synthetic data.

2. Experimental Validation on Cylinder Flow

- Use surface sensors to reconstruct vortex shedding.
- Evaluate latency, robustness, noise handling.
- Benchmark predictions versus reference.

Deliverables

- Working DA pipeline.
- Paper: Real-time estimation of vortex shedding dynamics from surface sensors using data assimilation.

PHASE 2 (Year 2): Main Contribution — Airfoil With Dynamic Stall

Goals

- Demonstrate robust real-time prediction of a dynamically changing aerodynamic state.
- Produce the core results of the PhD.

Key Actions

1. Design and Execute Dynamic Stall Experiments

- Airfoil with controlled pitch ramps or oscillations.
- Distributed surface sensors.
- Force measurements and optional flow field validation.

2. Real-Time State Estimation

- Prediction of lift, moment, flow regime transitions, and approximate separation.
- Evaluate latency relative to flow timescales.

3. Model Refinement

- Improve ROM structure.
- Enhance DA stability under nonlinear transitions.

Deliverables

- Core dataset and results.
- High-quality journal paper.
- Majority of thesis chapters drafted.

PHASE 3 (Year 3): Transition to High-Speed Wind Tunnel Demonstration

Goals

- Adapt the method to compressible/high-Re conditions.
- Conduct first high-speed wind tunnel campaigns.

Key Actions

1. Method Adaptation

- Incorporate compressibility effects as needed.
- Sensor calibration for high-speed conditions.
- Robustness testing.

2. Experimental Campaign #1

- Reproduce simplified separated flow.
- Validate real-time estimation with surface sensors only.

Deliverables

- High-impact journal paper.
- Proof-of-concept demonstration in relevant aerospace conditions.

PHASE 4 (Year 4): Final Demonstration and High-Speed Dynamic Stall / Buffet

Goals

- Execute an ambitious high-speed experiment demonstrating full method capability.
- Deliver final synthesis of framework and thesis.

Key Actions

1. Full-Scale Demonstration

- Dynamic stall or moderate transonic buffet.
- Real-time prediction of aerodynamic forces and flow regime transitions.

2. Final Synthesis

- Integrate all components into a generalizable DA framework.
- Produce final thesis and publications.

Deliverables

- Final PhD thesis.
- Capstone publication.
- Dataset and software framework.

Scientific Impact

- Advances experimental data assimilation for nonlinear unsteady aerodynamic flows.
- Provides a framework for real-time reconstruction using minimal sensing.

Technological Impact

- Enables applications in maneuver load prediction, stall/buffet detection, aeroelastic monitoring, and flow control.

Strategic Impact

- Positions the group as a leader in real-time experimental DA for aerodynamic systems.